



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Computer Science and Engineering
Academic Year: 2025-26 (semester I)
In-Semester Examination II

Class: SY	Course Name: Digital Electronics	Course Code: PCC-CS203
Date: 13.10.2025	Time: 14:30 to 15:15	Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.			Marks	CO	BT
Q.1	a	Define JK and D flip-flop with truth tables.	02	CO1	BT1-BT2
	b	Explain how race-around condition occurs in JK flip-flop and how it is avoided.	03	CO1	BT1-BT2
Q.2	a	Convert the SR to JK flip-flop and derive the excitation table .	02	CO3	BT2-BT3
	b	Convert the JK to T flip-flop and derive the excitation table .	03	CO3	BT2-BT3
Q.3		Design a Mod-5 asynchronous counter using T flip-flops . Draw the circuit diagram and explain its working sequence.	05	CO4	BT3-BT4
Q.4		Design a 4-bit synchronous ring counter using T flip-flops that follows state sequence: 1000→0100→0010→0001→1000.	05	CO5	BT4-BT6

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Explain the number systems, basic logic gates. Boolean algebra, and logic realizations using switching functions using gates.
CO 2	Understand the CAD of logic circuits through VHDL and FPGAs, CPLDs.
CO 3	Understand the design and analyse combinational and sequential logic circuits.
CO 4	Understand the basic software tools for the design and implementation of digital circuits and systems.
CO 5	Reinforce learned principles and techniques taught in the classroom through theory assignments and build projects for the society.